



COSMO INSTRUMENTS (I) PVT. LTD.

SPECIALIST IN AIR LEAK AND FLOW TEST



AIR LEAK TESTER

MODEL LS-1864X

ERROR AND TROUBLESHOOTING MANUAL

1 Error Messages

Errors are detected in either the power-on check procedure, or during execution of a leak test. If an error is detected, error code will be displayed on the leak rate indicator along with blinking No-Go LED and beep sound. An ERROR signal will be transmitted through the control I/O port. Pressing **STOP** key clears the error display and signal.

1.1 Errors and Treatments

No.	Display	Probable Cause	Treatment
1	E-1	Flash memory error	Replace the CPU board.
2	E-2	DPS stopped oscillating.	Sensor electric source, cable disconnection, electric circuit damage. Contact Cosmo for repair.
3	E-3	Excessive DPS zero-point shift	Check and adjust the stop zero point. (Refer to 12.4.2)
		Water, oils, or other contaminants in DPS	After checking and adjusting the stop zero point, repeat the charge hold to check the zero point several times. If the stop zero point tends to shift or in case of no repeatability, DPS may be contaminated. Replace the DPS. (Repair by Cosmo) Judged in DL1 stage.
4	E-4	Improper test pressure	Test pressure may be out of range of the pressure switch. Check the test pressure setting. Judged at the end of Pressurization (CHG)
5	E-5	Air Operated Valve Error	Check the pilot pressure. Adjust the pilot pressure between 400 kPa and 700 kPa. (Refer to 12.4.4) Judged in CHG or DL3.
6	E-6	A/D Converter error	Replace the CPU board.
7	E-7	DPS output saturated	Judged at the end of DET stage. No-Go judgment will be made. ERROR signal won't be transmitted.
8	E-8	I/O output error	Cable disconnection, electric circuit damage. Contact Cosmo for repair.
9	E-9	Pressurization valve error	Pressurization valve AV1 does not close. Air operated valve AV1, pilot pressure solenoid valve error. Contact Cosmo for repair.

The LO NO GO LED will light when a static memory (SRAM) error is detected. In this case, the CPU board must to be replaced.

1.2 E-3: Excessive Shift of DPS Zero-point

When a leak test is repeated without unclamping after the leak test or Charge-hold, excessive shift DPS zero-point E-3 may occur.

This tends to occur when the volume of the tested part is so large or the piping is so narrow that the air inside the tested part is not completely exhausted, resulting remaining pressure is detected during DL1 stage in the following test.

In this case, delay the input of the next START signal or extend the DL1 timer to let the tester exhaust completely.

The monitoring function may also be disabled with a memory switch. (Refer to 10.2.3)

1.3 E-5: Air Operated Valve Error

When the error E-5 is indicated, check the followings. (Refer to 4.4)

1) **Test pressure**

A positive pressure of 8 kPa or more or a negative pressure of -13 kPa or more is required.

2) **Pilot pressure**

Adjust the pilot pressure between 400 kPa and 700 kPa.

3) **DL3 timer**

A Low test pressure model will require longer DL3 timer than usual. Extend the DL3 timer by 1 to 2 seconds. In case of Micro-low pressure model, it may be required to replace the orifice.

4) **Air-operated valves and solenoid valves**

If the valves are contaminated, they don't operate. Cleaning or replacement is required.

5) **DPS sensitivity**

Check the DPS sensitivity using a Leak Master, and DPS span.

1.4 E-9: Fill Valve Error

In case there is a large difference between the WORK and MASTER volumes or it takes a longer time for exhaust, the offset of the DPS could exceed its limit and display E-9 error message. In this cases, set 5 sec or longer time with the memory switch. (Refer to 10.2)

2 Leak Test Troubleshooting

2.1 When No-Go Occurs Frequently

No-Go judgment	Category	Probable Cause	Treatment
No-Go occurs frequently for Work side	Tested part	Internal mold cavity of the tested part. Leaks from sealing caused by machining defects.	Perform charge-hold applying soap solution or water-immersion test on the fixture.
		Differential pressure due volume change caused by part and/or internal fixture deformation.	Change the timer settings or test pressure.
		Differential pressure due to temperature changes.	Let the part to be ambient temperature.
	The sealing fixture, piping or other parts	Worn-out or contaminated sealing rubber	Clean or replace sealing rubber as required.
		Leak from the piping or valves.	Perform charge-hold applying soap solution on the piping and valves.
		Differential pressure due to seal volume changes caused by clamping pressure.	Implement exhaust interference prevention to guard against effects from other testers if any. Review the sealing fixture and rubber designs.
		Temperature changes of the fixture after water-immersion	Blow the water away and control the water temperature.
	Tester	Internal leak or contamination of DPS	Perform No Leak test with the stop valves closed.
		Improper test timer	Execute Master-Preset sampling with non-leaking part, and check if the value is within the limit. Change the timer settings.
	Master-Preset Compensation	Improper equivalent internal volume Improper tested part (leak, deformation or temperature effects) The part used for Master-Preset value sampling is not actual part.	Use actual non-leaking tested part for sampling Master-Preset value.
When the BAL stage completes, the No Go occurs.		Large leak	After checking the sealing rubber for wear or stuck foreign matter, replace with a conforming tested part.
		Improper test pressure	Check the pressure source. Check the regulator setting.

No-Go judgment	Category	Probable Cause	Treatment
The Master-Presetting results in No Go judgment.		Leak from the tested part, etc.	Retry Master-Preset with another part.
		Temperature change in the tested part	Do not carry out Master-Preset with a special part, such as one that has just been subjected to water-immersion checking.
No-Go occurs frequently for Master side.		Leak from the master	Replace the master.
		Poor master temperature stability	Replace with a Master Chamber having good temperature stability.

3 Locating Leaks in Places Other Than the Leak Tester

- a) Press the CHG HOLD key to pressurize the tested part and locate leaks applying soap solution.

NOTE: Do not apply soap solution to a vacuum pressure type tester. Disconnect the piping and fixture from the WORK-side port and apply regulated micro-low pressure and soap solution to locate the leak.

NOTE: If soap solution is used, be sure to wipe it off with a cloth.

- b) Switching the master and tested part to perform leak test

If a tested part was rejected but the leak can't be located even by applying soap solution, switch the tubing and fixtures of the MASTER-side and WORK-side. This helps find out whether the leak is from the MASTER-side or WORK-side piping or fixture.

If the polarity of the leak rate changes when piping and fixtures of the MASTER and the WORK are switched, a leak is more likely to locate in the tested part or fixture seal. Do this check with the Master-Preset feature disabled.

Be sure to replace the piping in original position when the check is completed.